# Autonomous Rover Terrain-Risk Classification

*A Zero-Leakage Machine Learning Pipeline for IMU-Based Navigation Risk Prediction*

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**CS3006: Introduction to Data Science for Engineers**


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# Abstract

Autonomous rovers must adapt their navigation parameters based on the terrain to prevent mechanical stalling and loss of traction. This project develops a zero-leakage machine learning pipeline to classify 128-sample Inertial Measurement Unit (IMU) windows into three navigation-risk tiers: Firm, Compliant, and Irregular. Using the Kaggle CareerCon 2019 dataset, we evaluated Logistic Regression, Random Forest, and a PyTorch 1D Convolutional Neural Network (CNN). By implementing a strict session-grouped cross-validation protocol, we prevented the artificial 87% Macro-F1 score inflation caused by data leakage. Our final CNN model achieved a hazard recall of 78.1% on unseen test sessions. Error analysis revealed that IMU sensors struggle to differentiate between smooth firm surfaces (wood) and smooth compliant surfaces (carpet) due to similar vibration profiles, providing a clear physical diagnosis of the model's limitations.


# 1. Introduction

For autonomous rovers operating in unmapped or mixed indoor environments, the physical properties of the floor directly dictate safe operating parameters. Driving at full speed over a compliant surface, such as carpet, drastically increases rolling resistance and risks motor stalling. Conversely, driving over tile gaps risks severe jolts and loss of traction.

The objective of this project is to build a robust classification system that relies solely on internal IMU data (accelerometer and gyroscope) to predict terrain stability. The predictions are mapped to actionable navigation-risk tiers.

# 2. Dataset and Problem Formulation

## 2.1 Data Source and Structure

The dataset is derived from the Kaggle "CareerCon 2019 – Help Navigate Robots" competition. It contains 487,680 rows of sensor data, formatted as 3,810 distinct time-series windows. Each window consists of exactly 128 sequential samples across 10 channels (orientation, angular velocity, and linear acceleration). The data spans 73 continuous recording sessions across 9 floor surfaces.

## 2.2 Navigation-Risk Tiers

Rather than classifying the exact material, surfaces were grouped by their required robotic control response:

- **Tier 0 (Firm):** Concrete, tiles, wood. Safe for full-speed operation.
- **Tier 1 (Compliant):** Carpet, soft PVC. High drag, sinkage risk. Requires high torque and lower speed.
- **Tier 2 (Irregular):** Tile gaps. Traction loss risk. Requires damping and reduced speed.

# 3. Methodology

## 3.1 Data Preprocessing and Feature Engineering

To ensure the model learns terrain dynamics rather than session-specific biases (like the starting direction of the robot), absolute heading data was dropped. We engineered 149 features based on physical vehicle dynamics:

- **Gravity Projection:** Linear acceleration was split into vertical (vibration and bumps) and horizontal (drag and slip) components.
- **Spike Isolation:** A centered rolling mean (window = 5) was subtracted from the raw signal to isolate high-frequency jolts from the robot's slow, intentional movements.
- **Spectral Features:** Band-power fractions and dominant frequencies were extracted to capture the periodic nature of tile joints.

Extreme outliers (robust z-score > 6) were retained, as large impacts represent genuine physical signals rather than sensor noise. Features were standardized and winsorized using parameters fitted strictly on the training set.

## 3.2 Zero-Leakage Splitting Strategy

A critical flaw in standard random splitting is data leakage. Time-series windows from the same recording session share identical background noise. If windows from the same session appear in both training and test sets, the model learns to identify the session rather than the terrain.

We utilized a **Stratified Group K-Fold** split based on `group_id` (session ID). 43 sessions were used for training, 15 for validation, and 15 strictly held out for final testing.

## 3.3 Model Architectures

Three models were evaluated:

1. **Logistic Regression:** A regularized linear baseline to test the efficacy of the engineered features.
2. **Random Forest:** A non-linear ensemble (tuned to 200 trees, max depth 8) robust to heavy-tailed sensor spikes.
3. **PyTorch 1D-CNN:** A deep learning model trained directly on the raw signal sequences. It utilizes 3 layers of Conv1d-BatchNorm-ReLU, global pooling, and dropout. To prevent overfitting on the small dataset, it was constrained to 7,299 parameters.

# 4. Results and Evaluation

Because the dataset is heavily imbalanced (60% of the data belongs to the safe "Firm" tier), accuracy is a misleading metric. Our primary evaluation metrics are **Hazard Recall** (the percentage of actual hazards successfully detected) and **Macro-F1**.

### 4.1 Model Comparison on Unseen Test Sessions

**Table 1:** *Performance metrics on 15 held-out test sessions.*

| Model | Macro-F1 | Hazard Recall | Hazard Precision | Balanced Acc. |
|---|---|---|---|---|
| Majority Baseline | 0.250 | 0.000 | 0.000 | 0.333 |
| Logistic Regression | 0.452 | 0.704 | 0.576 | 0.471 |
| Random Forest | **0.478** | 0.734 | **0.642** | **0.504** |
| **PyTorch 1D-CNN** | 0.473 | **0.781** | 0.630 | 0.503 |

*Best value per metric in bold; final model highlighted.*

The 1D-CNN achieved the highest hazard recall (78.1%), successfully outperforming the classical machine learning models in identifying dangerous terrain.

### 4.2 Error Analysis and Physical Diagnosis

Despite beating the baselines, the overall error rate was 32%. An in-depth error analysis provided strong evidence for the limitations of IMU-only classification:

- **Concentrated Errors:** 58% of all misclassifications originated from just 3 of the 15 test sessions.
- **The Wood Confounder:** 88 out of 92 errors on wood floors were incorrectly predicted as “Compliant” (soft).
- **Evidence:** The median peak rotation rate on wood floors was 0.185 rad/s, which closely mirrors the 0.165 rad/s of soft carpets. Because an IMU measures vibration rather than actual material stiffness, a smooth firm floor feels identical to a smooth soft floor.

## 5. Conclusion

This project successfully implemented an end-to-end, zero-leakage pipeline for autonomous rover terrain classification. We showed that random data splitting artificially inflates Macro-F1 scores from 0.65 to 0.87 due to session memorization. While the PyTorch 1D-CNN effectively captures temporal dynamics to detect 78.1% of terrain hazards, our evidence shows that IMU sensors alone are insufficient for perfect stall prevention. Future work should focus on sensor fusion, integrating wheel odometry (to detect physical slip) and depth cameras (to predict terrain compliance visually) to resolve the physical ambiguities identified in this study.